



CS253 PROJECT

CMMS

Centralized Mess Management System

A comprehensive full-stack web application for managing daily mess operations, billing, and student interactions at IIT Kanpur



GitHub Repository

github.com/Chronos193/CMMS

 IIT Kanpur

 Full-Stack Web Application



THE CHALLENGE

Need for the Software



Manual Menu Viewing

Students must physically visit the mess to check daily menus, causing inconvenience and wasted time.



No Digital Extras Booking

No digital system exists for students to book add-on items, leading to long queues and inventory issues.



Paper-Based Rebates

Rebate applications require physical forms with lengthy processing times and no tracking mechanism.



Opaque Complaint System

Complaint resolution lacks transparency with no way for students to track status or receive updates.



Manual Billing Errors

Billing calculations are done manually, resulting in frequent errors and disputes with students.



No Centralized Communication

No unified platform exists for admins to notify students about important updates or changes.

THE SOLUTION

Solution Overview

CMMS digitizes the **entire lifecycle** of mess management — from daily menu viewing to extras booking, rebate applications, complaint resolution, automated billing, and admin oversight.

Potential Users



Students

Residents of IIT Kanpur halls (Hall 1-13, GH1) who need mess services and want digital convenience



Mess Administrators

Staff who manage operations, inventory, billing, and student communications



14

Data Models

37

API Endpoints

18

Page Routes

Architecture & Tech Stack

Decoupled SPA Architecture



Frontend

React 19 + Vite 8 + Tailwind CSS 4

Vercel



Backend

Django 6 + DRF

Railway



Database

PostgreSQL (Prod) / SQLite (Dev)

Authentication

JWT tokens stored in HttpOnly cookies with CSRF protection for authentication

Backend Stack

Django 6.0

Web Framework

DRF

API Layer

SimpleJWT

Authentication

ReportLab

PDF Generation

Brevo API

Email Service

Poetry

Dependency Mgmt

Frontend Stack

React 19

UI Framework

Vite 8

Build Tool

Tailwind 4

Styling

Framer Motion

Animations

html5-qrcode

QR Scanning

Axios

HTTP Client

LIVE PRESENTATION

Demo Time!

Live demonstration of CMMS application showcasing student
and admin workflows



Student Portal



Admin Dashboard

Future Development Plans



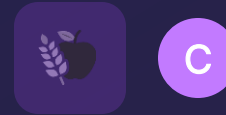
Mobile App

Develop native iOS and Android apps for better accessibility, push notifications, and offline capabilities



Payment Gateway

Integrate UPI, credit cards, and net banking for direct bill settlement within the app



Nutrients Tracker

Real-time calorie and macronutrient (carbohydrates, proteins, fats) breakdown for daily mess menus.

Future Development Plans



Analytics Dashboard

Provide admins with insights on popular extras, peak ordering times, and complaint trends



Dark Mode

Add dark mode support to our website to give it a sleek and professional look and enhance the browsing experience.

Lessons Learnt



Hands-On Learning

This project provided practical experience in building a full-stack application with real-world constraints like concurrency control, security measures, and deployment challenges



Technical Depth

Gained deep understanding of JWT authentication with HttpOnly cookies, database transactions with `select_for_update()`, and decoupled SPA architecture



Testing Importance

Comprehensive pytest suite with 37 API endpoint tests taught the value of automated testing and test-driven development



Deployment Experience

Learned to deploy frontend (Vercel) and backend (Railway) separately with CORS configuration, environment variables, and production security settings

INTERACTIVE SESSION

Q&A

Questions & Answers



Thank You



Questions?



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IIT Kanpur



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